14 a and b are odd numbers.

(a) Give an example to show that the value of $2(a + b)$ is a multiple of 4

Choose $a = 3$ and $b = 5$. Both numbers are odd.

1 mark
Substitute two odd numbers

$2(a + b) = 2(3 + 5) = 16 = 4 \times 4$

1 mark
Show the result is a multiple of 4

(2)

(b) Show that, when a and b are both odd numbers, the value of $2(a + b)$ will always be a multiple of 4

Odd + odd = even, so $a + b$ is even.

1 mark
Explain why a + b is even

An even number is $2k$ for some integer k.

Therefore $2(a + b) = 2 \times 2k = 4k$.

1 mark
Twice an even number is a multiple of 4

(2)

(Total for Question 14 is 4 marks)